A Design-Oriented Study of the Nonlinear Dynamics of Digital Bang–Bang PLLs

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***Abstract*—The use of bang–bang phase-locked loops (BBPLLs) has become increasingly common in a lot of communications systems, in particular in the area of clock and data recovery. Although most of the BBPLLs implemented up to now use analog loop filters, the binary output of the phase detector naturally lends itself to a digital implementation. In this paper, the nonlinear dynamics of first- and second-order digital BBPLLs is analyzed from a design perspective. In particular, the effects of loop delays on the PLL performances are emphasized. Conditions for the existence of orbits (limit cycles) are derived, and the timing jitter performances are evaluated. Finally, useful expressions for the design and optimization of the PLL parameters for low jitter are given.**

***Index Terms*—Bang–bang control, digital phase-locked loop (PLL), jitter, limit cycles, nonlinear systems, stability criteria.**

I. Introduction

IN RECENT years, great attention has been paid to a class of nonlinear phase-lock circuits called bang–bang phase-locked loops (BBPLLs). The particularity of this class of phase-locked loops (PLLs) is the use of a phase detector which quantizes the phase-difference information with one-bit resolution. Since the output can assume only two different values, this class of phase detectors is called binary phase detectors (BPDs). Other names used are bang–bang or lead–lag phase detectors. BBPLLs are already widely used in the area of clock and data-recovery circuits, due to the unique advantage of inherent sampling phase alignment [1], [2] and high-frequency capabilities. Although most of the BBPLLs implemented up to now use analog loop filters, the binary output of the phase detector lends itself naturally to a digital implementation. Advantages of the digital implementation are lower chip area, lower power consumption, robustness against technology parameter variations, and friendly realization in deep submicron and low-voltage technologies. For those reasons, increasingly, other areas of communications are looking with interest at BBPLLs as clock-multiplying units.

The presence of the BPD introduces a hard nonlinearity in the loop, invalidating the traditional Laplace domain analysis used in linear PLLs. The fundamental aspect of BBPLLs is the inevitable presence of limit cycles in the dynamics. Indeed, a BBPLL cannot lock to the reference clock in a traditional sense, where the output of the phase detector and the loop filter voltages settle asymptotically around a fixed value, disturbed only by thermal noise. The locked state of the BBPLL is practically described by an orbit in an appropriate phase plane. With the term orbit, we intend a trajectory on the phase plane which repeats itself (the difference between limit cycle and orbit is a minor one, and, although limit cycle is a more common term, we will use the term orbit for correctness).

The condition of stability for a BBPLL translates then in the condition for the existence of orbits in the nonlinear dynamical behavior.

An approach used for the analysis of BBPLL is based on the linearization of the system and the subsequent use of linear techniques [3]. Also in this direction, an equivalent bandwidth concept for BBPLLs has been proposed in [4] which could help in the analysis of the jitter transfer function. Despite giving a good insight into the dynamics of the system, this approach cannot detect the presence of orbits.

Recent works have studied the nonlinear dynamics of a class of digital PLLs showing frequency quantization due to the presence of a numerically controlled oscillator [5], [6]. These studies demonstrated that the theory of nonlinear control can be effectively applied to the analysis of the PLL dynamics, confirming the results that were obtained relying on simulations [7] and providing also indications of other features of the system behavior.

The goal of this paper is the evaluation of the nonlinear dynamics of a digital second-order BBPLL. The approach used here is design oriented, sometimes sacrificing the mathematical rigor.

In particular, it is of interest to determine the behavior of the BBPLL in the locked state, and derive a useful expression for the estimation of timing jitter. Since we are interested here only in the quantization noise produced by the hard nonlinearity in the loop, it will be assumed that all the PLL building blocks are free from any kind of physical noise source. Special attention is paid to the impact of the loop delays on the dynamics of the system.

The paper is organized as follows. The BBPLL architecture under study is presented in Section II. In Section III, a simplified nonlinear model for the BBPLL is derived. Section IV briefly analyzes the case of a first-order BBPLL. Since this architecture is rarely used in practical implementations due to a reduced PLL frequency-capture range [3], the paper concentrates on the analysis of second-order BBPLLs. This is done in Section V, where analytical conditions for the existence of orbits, hence, for the stability of the BBPLL, are derived. Expression for jitter estimation are derived and, finally, design guidelines for the minimization of the jitter are provided. Section VI will show simulation results which confirm the previous analysis.

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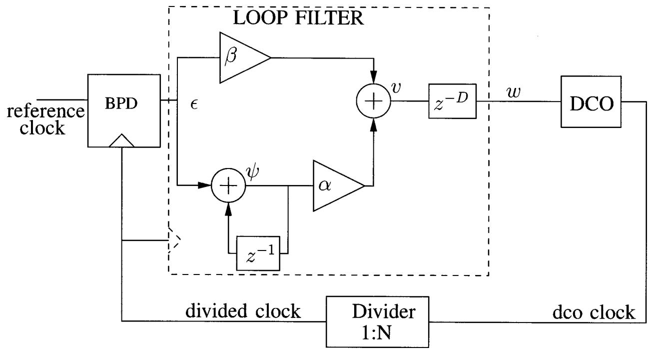

Fig. 1. Block diagram of a second-order digital BBPLL.

II. BBPLL Architecture

The block diagram of the digital BBPLL under consideration is reported in Fig. 1. The PLL consists of a BPD, a digital loop filter, a digitally controlled oscillator (DCO), and a feedback divider. The function of the BPD is to provide an indication of the relative phase difference between the reference clock and the feedback clock in a binary form. Its operation is logically identical to the operation of an ideal sampling register, with the reference clock as data input and the divided clock as sampling clock. The binary phase information is fed to the digital loop filter, which consists of a proportional and of an integral path. The constants in the two paths will be indicated with β and α for the proportional and integral paths, respectively. The digital filter is usually clocked by the divided clock itself, which, in locked conditions, is synchronous to the reference clock. In a lot of applications where the BPD is nowadays used, the reference frequency is quite high, ranging from hundreds of megahertz to some gigahertz. Therefore, the operations to be performed by the loop filter may require more than one clock cycle of the reference clock. The eventual pipeline stages introduced in the actual implementation of the PLL are modeled by the z^{-D} block, with D the number of reference clock delays. In a straightforward implementation of the DCO, a digital-to-analog converter converts the digital loop filter information into an analog voltage, which in turn drives the control line of a voltage-controlled oscillator. The description of other possible implementations of the DCO is not in the scope of this paper.

III. BBPLL Nonlinear Model

The classical treatment of linear PLLs is done in the frequency domain by use of the Laplace transform. Due to the presence of the nonlinear BPD block in the loop, this approach cannot be used for the BBPLL. The analysis presented in this paper is carried out completely in the time domain, allowing an easy mathematical modeling of all the blocks in the PLL loop.

The logical binary output value of the BPD is mapped to a numerical value $\epsilon = 1$ if the divided clock leads the reference, or to $\epsilon = -1$ if the reference leads the divided clock. By indicating with t_{r} and t_{d} the time instants of the rising edges of the reference and divided clocks, respectively, we can write:

$$\epsilon = \mathrm{sgn}(\Delta t)$$

where $\Delta t \triangleq t_{\mathrm{r}} - t_{\mathrm{d}}$, and sgn is the sign function defined as follows:

$$\mathrm{sgn}(x) = \begin{cases} 1, & \text{if } x \geq 0 \\ -1, & \text{if } x < 0. \end{cases}$$

Note that Δt indicates the timing error between the reference clock and the divided clock. Assuming a noise-free reference clock, Δt represents also the jitter of the PLL. This model of the BPD does not account for the presence of the duty cycle in the reference clock. Hence, it cannot be used to describe the locking behavior of the BBPLL in presence of cycle slips.

The DCO is modeled as a linear block with the period of the output clock (T_v) depending linearly on the input control value w, according to the following:

$$T_v = T_{v0} + K_T \cdot w$$

where T_{v0} is the free-running period, and K_T is the period gain constant. In this model, the response of the DCO to a change of w is immediate without any transient behavior. In other words, we assume a DCO time constant which is negligibly smaller than the reference clock period. This assumption is well satisfied for most DCO implementations. Although modeling the oscillator as a frequency-controlled oscillator is the traditional way of analyzing analog or mixed-mode PLLs, in this case, it is more convenient to approach the analysis in the time domain. The two approaches are equivalent, provided $K_T = -K_F \cdot T_{v0}^2$, where K_F is the DCO frequency gain (see also [8]).

The next step is to find the set of equations describing the dynamics of the system. As the values of the quantities ϵ, ψ, and v change only on the rising edges of the divided clock, it is possible to describe the system with a set of finite-differences equations. In the following, the subscript k on the symbols indicates the value at the kth sampling instant (corresponding to the kth rising edge of the divided clock).

Fig. 2 represents the flow of the signals of the BBPLL. By inspection of Figs. 1 and 2, the dynamics of the BBPLL can be described by the following set of equations ($T_{r,k}$ indicates the kth period of the reference clock):

$$\begin{cases} \Delta t_k = t_{\mathrm{r},k} - t_{\mathrm{d},k} \\ \epsilon_k = \mathrm{sgn}(\Delta t_k) \\ \psi_k = \psi_{k-1} + \epsilon_k \\ v_k = \beta\epsilon_k + \alpha\psi_k \\ T_{v,k} = T_{v0} + K_T \cdot v_{k-D} \\ t_{\mathrm{r},k+1} = t_{\mathrm{r},k} + T_{r,k} \\ t_{\mathrm{d},k+1} = t_{\mathrm{d},k} + NT_{v,k}. \end{cases} \tag{1}$$

By subtracting the last two equations, an expression for Δt_{k+1} as a function of Δt_k can be found, thus eliminating t_{r} and t_{d} from the equation set (1):

$$\Delta t_{k+1} = \Delta t_k + T_{r,k} - NT_{v0} - NK_T v_{k-D}.$$

The system can be described using only two state variables, namely Δt and ψ. As the goal of this paper is the analysis of the BBPLL dynamics in the presence of a nonmodulated and noise-free reference clock, we can drop the dependence of T_r on

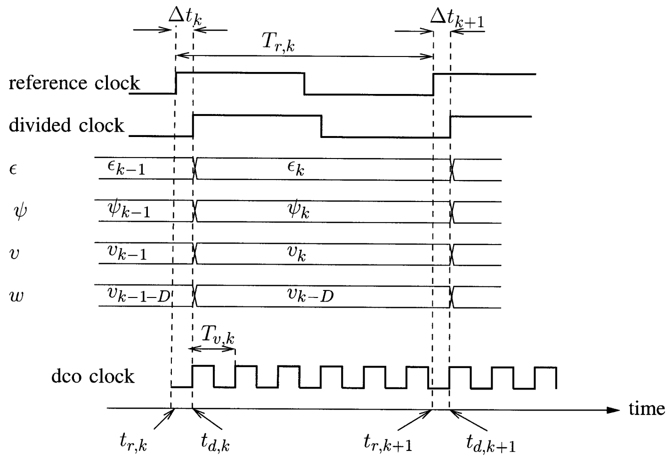

Fig. 2. Time diagrams of the signals in the BBPLL.

k. Elimination in (1) of ϵ, v, and T_v through substitution leads to

$$\begin{cases} \Delta t_{k+1} = \Delta t_k + T_r - NT_{v0} - N\alpha K_T \cdot \psi_{k-D} - \cdots \\ \qquad - N\beta K_T \operatorname{sgn}(\tau_{k-D}) \\ \psi_{k+1} = \psi_k + \operatorname{sgn}(\Delta t_{k+1}). \end{cases} \tag{2}$$

The definition of the following quantities:

$$\begin{aligned} \tau &\triangleq \frac{\Delta t}{N\beta K_T} \\ x_0 &\triangleq \frac{T_r - NT_{v0}}{N\beta K_T} \\ R &\triangleq \frac{\alpha}{\beta} \end{aligned} \tag{3}$$

allows the set of equations to be rewritten in the following simplified form:

$$\begin{cases} \tau_{k+1} = \tau_k + x_0 - R \cdot \psi_{k-D} - \operatorname{sgn}(\tau_{k-D}) \\ \psi_{k+1} = \psi_k + \operatorname{sgn}(\tau_{k+1}). \end{cases} \tag{4}$$

This is the nonlinear map ruling the evolution of the BBPLL in time. Some words on the definitions (3) are in order. βK_T and $N\beta K_T$ are the quantization steps of the DCO and of the divided clock period, respectively, due to the proportional branch of the BBPLL. τ is the timing jitter normalized to this quantization step. The quantity x_0 is the normalized difference between the DCO free-running period multiplied by N and the reference clock period. It is zero only if the two periods are identical. Due to the normal tolerances of the analog circuit components, this condition can never be met in a practical BBPLL implementation. Nevertheless, the assumption $x_0 = 0$ can be used as a starting point in order to simplify the analysis and get a clearer insight into the nonlinear dynamics of the system. Finally, R represents the ratio of the integral to the proportional constants of the digital loop filter.

A block diagram representing the simplified nonlinear model (4) is shown in Fig. 3.

IV. Analysis of First-Order BBPLL

In this section, the special case of a BBPLL without an integral branch in the loop filter (first-order BBPLL) is analyzed.

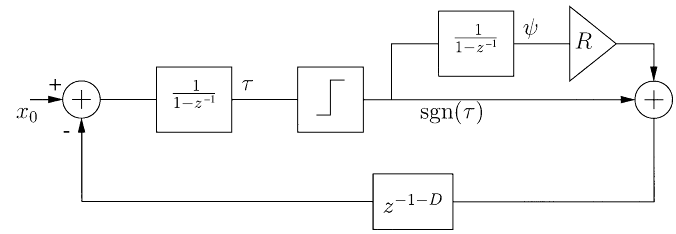

Fig. 3. Simplified nonlinear model for the second-order BBPLL.

Although this architecture is rarely used in practical implementations, due to a reduced PLL frequency-capture range, it will give useful insights into the nonlinear dynamics. In this case, the nonlinear map (4) reduces to

$$\tau_{k+1} = \tau_k + x_0 - \operatorname{sgn}(\tau_{k-D}). \tag{5}$$

Interestingly enough, with $D = 0$, this equation reduces to the extensively studied nonlinear model of the first-order $\Delta\Sigma$ modulator using a one-bit quantizer (see, e.g., [9]). While in those studies the focus is on the statistical properties of the quantizer output, for the BBPLL case, the input of the quantizer (τ) is of interest. Indeed, the statistics of τ determines the jitter performance of the BBPLL, while the output of the quantizer is just an internal BBPLL signal.

The analysis will be performed in two steps. In the first step, input x_0 will be assumed equal to zero. This assumption will be removed in the second step.

A. Case $x_0 = 0$

Equation (5) simplifies to

$$\tau_{k+1} = \tau_k - \operatorname{sgn}(\tau_{k-D}). \tag{6}$$

The presence of $D \neq 0$ in the map increases the order of the system from a point of view of the nonlinear dynamics. In order to specify a unique trajectory, a single initial condition is not sufficient, since the sign of the previous D iterates will also affect the trajectory. However, a closer analysis shows that the only possible steady-state trajectories are those showing a triangular shape of τ_k as a function of k. A simple argument can be used to demonstrate that. First of all, an inspection of (6) reveals that the trajectory of τ will move with unity steps either in the positive or negative direction. The trajectory can not stay indefinitely in the positive or negative axes, since the term $-\operatorname{sgn}(\tau_{k-D})$ will push it to the other side. Hence, at some point in time, τ will assume a value τ_0 inside the interval [0,1). Assume that the trajectory has its maximum positive value at $\tau_0 + N$, with N a non-negative integer. This means that at least $2N + 1$ iterates around this point have a positive sign, and sooner or later (it depends on the value of D), they will push the trajectory in the negative direction for $2N + 1$ consecutive steps. If this sequence of consecutive negative steps starts when the trajectory has the value $\tau_0 + N - 1$ (or lower), after $2N + 1$ negative steps, the trajectory will reach the point $\tau_0 - N - 2$ (or lower). That means that now $2N + 3$ (or more) iterates around this point have a negative sign. The same argument can now be applied to show that there will be at least $2N + 3$ consecutive steps in the positive direction, leading to a positive value of the trajectory higher than the assumed maximum value $\tau_0 + N$. The only way to have a

TABLE I
TRAJECTORIES OF FIRST-ORDER BBPLL FOR DIFFERENT DELAY VALUES

Iterate k	$D=0$	$D=1$	$D=2$
0	τ_0	τ_0	τ_0
1	τ_0-1	τ_0-1	τ_0-1
2	τ_0	τ_0-2	τ_0-2
3	·	τ_0-1	τ_0-3
4	·	τ_0	τ_0-2
5	·	τ_0+1	τ_0-1
6	·	τ_0	τ_0
7	·	·	τ_0+1
8	·	·	τ_0+2
9	·	·	τ_0+1
10	·	·	τ_0
11	·	·	·

steady-state trajectory is that the sequence of $2N+1$ negative steps starts exactly when the trajectory is at τ_0+N, leading to a triangular-shaped trajectory.

Among all possible triangular-shaped trajectories, the one with the maximum peak values has particular relevance. On the one hand, this trajectory leads to the maximum peak and root mean square (rms) jitter, a worst-case situation which must be within the design specifications. On the other hand, this trajectory corresponds to starting conditions $\tau_0 \in [0,1)$ and $\tau_{-1}\ldots\tau_{-D}$ all negative. The BBPLL will sooner or later run into this condition after startup, when it is not locked yet and the values of τ are most probably quite far from zero, either positive or negative. The following discussion then refers to the trajectories showing the maximum peak jitter. Equation (6) is simple enough to allow the trajectory of τ to be tabulated for different values of the delay D.

Table I reports the results for $D=0,1,2$. For the case $D=0$, the trajectory toggles between two points, τ_0 and τ_0-1. By increasing D, the trajectory expands its range both in the positive and in the negative direction on the τ axis, but it will always come back to the starting point τ_0 and repeat itself identically (this is indicated by the dots in the table). By extrapolating from the examples shown in Table I, some general properties of the trajectory can be deduced. First, the trajectory assumes values on the set $\{\tau_0+i, i=-D-1,\ldots,+D\}$ and, second, it is periodic with period $2(2D+1)$. That forms what is technically called a period-$2(2D+1)$ orbit. This information is enough to calculate the peak-to-peak jitter ($\tau_{\rm pp}$) as the difference between the maximum and the minimum values that τ assumes in the orbit $\tau_{\rm pp}=1+2D$.

An important observation is that each point of the set is visited by the trajectory twice per period, except the extremes, which are visited only once per period. The only exception is for $D=0$, when the orbit consists of only two points. That observation gives full information on the statistical distribution of τ and allows the easy calculation of τ's mean value μ_τ and jitter variance σ_τ^2

$$\mu_\tau = \tau_0 - \frac{1}{2}$$
$$\sigma_\tau^2 = \begin{cases} \frac{(1+2D)^2}{12}, & \text{for } D \neq 0 \\ \frac{1}{4}, & \text{for } D = 0. \end{cases}$$

The impact of the delay D on the performances of the first-order BBPLL is twofold. On the one hand, it worsens both peak-peak and rms jitter (σ_τ) in a linear way. One the other hand, it increases the period of the orbits, thus affecting the frequency distribution of the phase-noise spectrum of the BBPLL output. Indeed, for $D=0$, τ toggles between two values and the phase-noise spectrum of the BBPLL output will show a strong line (fundamental tone) at half the reference frequency $1/(2T_r)$ and at multiples of this frequency (harmonics). Higher values of D move down the fundamental tone toward lower frequencies and introduce other harmonics. In general, the fundamental tone will be placed at frequency $1/(2(1+2D)T_r)$. This could lead to problems if a BBPLL with a lot of pipelining is used in applications particularly sensitive to phase noise close to the carrier.

B. Case $x_0 \neq 0$

Inspection of (5) shows that if $x_0 \geq 1$, τ can not decrease with time, while if $x_0 \leq -1$, τ can not increase with time. Hence, the BBPLL can lock to the reference only if $|x_0|<1$. Remembering that x_0 is the normalized difference between the reference clock period and the DCO clock period, this condition is equivalent to that found in [3], and is practically the condition for no overload in $\Delta\Sigma$ first-order modulators with one-bit quantization. The dynamics described by (5) has been widely studied for the case $D=0$ (see, e.g., [9, Ch. 2]). One of the main results of this paper is that all trajectories are periodic when x_0 is rational, and quasi-periodic when x_0 is irrational. As the value of x_0 cannot be determined accurately, due to the natural tolerances in the circuits generating the reference and the DCO clocks, it make little sense to assume x_0 to be a rational number. Statistically, the probability that x_0 is rational is zero, while the probability that x_0 is irrational is one. In this section, we will assume x_0 to be irrational. For the same reasons expressed in the case of $x_0=0$, the analysis is limited to the orbits showing the maximum peak jitter. From (5), the maximum value that τ can assume is $(1+D)(x_0+1)$, and the minimum value is $(1+D)(x_0-1)$. The peak-peak jitter is thus $\tau_{\rm pp}=2(1+D)$.

It has been demonstrated that, for $D=0$, τ uniformly fills the interval $[x_0-1, x_0+1]$. Simulations show that this is, in general, not true for $D\neq 0$, being the probability distribution of τ inside the interval $[(1+D)(x_0-1),(1+D)(x_0+1)]$ is nearly, but not exactly, uniform. In particular, some subintervals show a probability density which is twice that of the rest of the interval. Some investigations are necessary to understand the distribution accurately. However, for design purposes it is often enough to have a reasonable worst-case estimation of the jitter generated by the PLL. By assuming τ to be uniformly distributed in the interval, a worst-case estimate for the jitter variance can be derived

$$\sigma_\tau^2 = \frac{(1+D)^2}{3}.$$

This result is slightly bigger than what was obtained for $x_0=0$. It can be noted that increasing D worsens both the peak-peak jitter and the rms jitter in a linear way. Little can be said about the spectral distribution of the phase noise of the DCO output clock without a detailed analysis of the trajectory in the time domain.

V. Analysis of Second-Order BBPLL

The second-order BBPLL is described by the nonlinear map (4) and presents two state variables, τ and ψ. The trajectory of the system will thus be analyzed in the bidimensional phase plane (τ,ψ). For the second-order BBPLL, it is sufficient to analyze the case for $x_0=0$. Indeed, defining $f=x_0/R$ and replacing x_0 in (4), one obtains

$$\begin{cases}\tau_{k+1}=\tau_k-R\cdot(\psi_{k-D}-f)-\mathrm{sgn}(\tau_{k-D})\\ \psi_{k+1}=\psi_k+\mathrm{sgn}(\tau_{k+1}).\end{cases}$$

Now subtracting f from both members of the second equation and replacing $\psi-f$ with ψ' in the previous set of equations, the nonlinear map reduces to (4) for the case $x_0=0$. Therefore, a system with $x_0\neq 0$ is described by the same nonlinear map as a system with $x_0=0$.

For $x_0=0$, the map (4) reduces to

$$\begin{cases}\tau_{k+1}=\tau_k-R\cdot\psi_{k-D}-\mathrm{sgn}(\tau_{k-D})\\ \psi_{k+1}=\psi_k+\mathrm{sgn}(\tau_{k+1}).\end{cases}\tag{7}$$

In the next subsections, the dynamics of this nonlinear map will be studied.

A. Phase Plane and Trajectories of the BBPLL

We will start the analysis by looking at the form of the trajectories of the system in the (τ,ψ) phase plane. A qualitative understanding of the behavior of the trajectories will allow us to derive analytical conditions for the existence of orbits, hence, for the stability of the BBPLL. Once the possible orbits are known, such properties as peak-peak jitter and jitter variance can be evaluated. Some important properties of the trajectories can be deduced by inspection of (7). The second equation of the set indicates that for positive values of τ, ψ increases by one after each iterate, independent of the value of τ. Conversely, for negative values of τ, ψ decreases by one after each iterate, independent of τ. Therefore, the trajectories plotted in a (τ,ψ) phase plane will move upwards on the right half-plane and downwards in the negative half. In a practical implementation of the loop filter, ψ will be represented by the value of a bus signal, which can be interpreted as a signed integer. Without loss of generality, then, it is assumed that $\psi\in\mathbb{Z}$, where $\mathbb{Z}$ is the set of signed integers.

The first equation of (7) indicates that the variation in the values of τ is the sum of two terms. One term $(-\mathrm{sgn}(\tau_{k-D}))$ depends only on the sign of τ at the Dth previous iterate and pushes the trajectory toward the axis $\tau=0$, as in the first-order BBPLL. The other term $(-R\cdot\psi_{k-D})$ depends linearly on the value of ψ at the Dth previous iterate, and can push the trajectory in the same or in the opposite direction as the first term. It can be noted that for $|\psi|\gg 1/R$, the second term is dominant over the first one.

Fig. 4 illustrates the direction of the trajectories on the (τ,ψ) phase plane for the case $D=0$ by use of a vector field. The vector field is symmetrical around the origin, and forms a sort of whirl around the origin. As for the case of the first-order BBPLL, if $D\neq 0$, the order of the dynamical system increases, and it is not possible to geometrically describe the dynamics in a bidimensional phase plane. Each trajectory is specified not only by the present values of τ and ψ, but also by their previous D iterates values. As the focus is to derive conditions for the stability of the system and to evaluate the worst-case jitter performances, orbits are considered which give rise to the maximum span on the τ axis. Looking at the nonlinear map (7), the maximum elongation of the trajectory on the τ axis, independent of the particular value of ψ, is obtained under the condition that when the trajectory crosses the axis $\tau=0$ from the positive (negative) to the negative (positive) half-plane, the previous D iterates have all positive (negative) values of τ. This is the same assumption done for the first-order BBPLL analysis and will be referred to as the maximum orbit assumption.

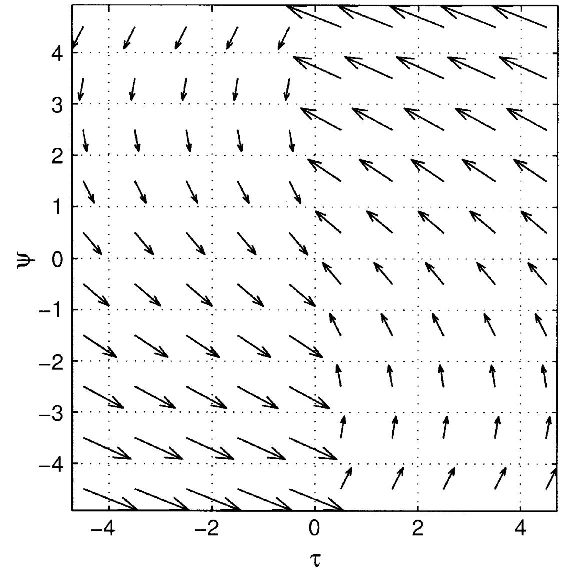

Fig. 4. (τ,ψ) phase plane with indications of the trajectory directions for $D=0$.

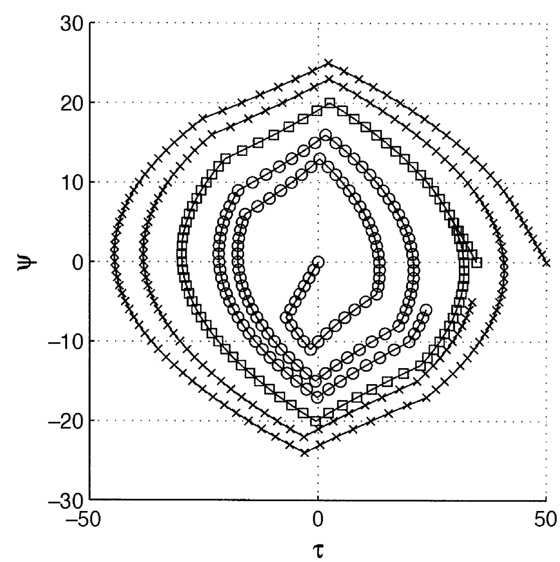

Fig. 5. Three examples of trajectories in the phase plane for $D=6$, $R=2/15$. $\psi=0$ and $\tau_0=0$ (circle), 34.66 (square), and 50 (cross).

Fig. 5 shows three examples of trajectories in the phase plane. The trajectories have been obtained with $D=6$, $R=2/15$, and three different initial conditions: $\psi=0$ and $\tau_0=0$ (circle), 34.66 (square), and 50 (cross). By changing the initial conditions, the trajectories show three different behaviors. Initial

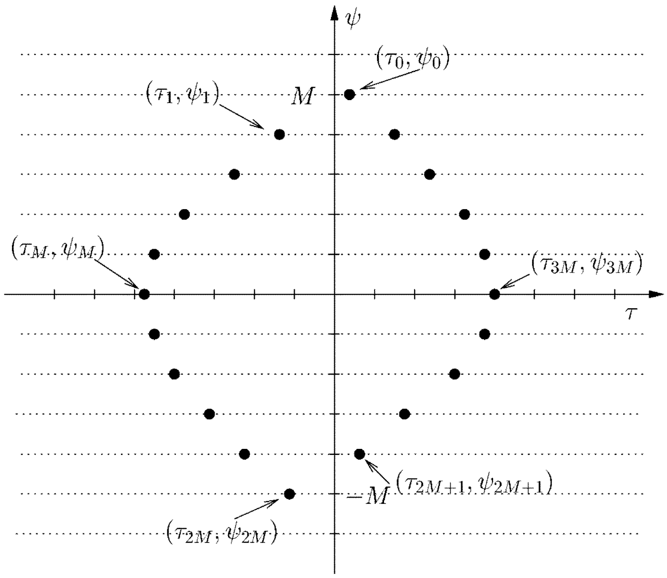

Fig. 6. Phase plane illustrating the points used to derive the condition for the existence of orbits in the second-order BBPLL dynamics.

condition (0,0) leads to a diverging trajectory. Initial condition (50,0) causes the trajectory to converge toward the origin. Initial condition (34.66,0) causes the trajectory to enter almost immediately a periodic orbit. This last case corresponds to the steady-state behavior of the BBPLL in locked condition.

B. Condition for the Existence of Orbits in the Dynamics of BBPLL

The conditions for the existence of orbits will now be derived. Fig. 6 illustrates a generic orbit used to derive those conditions. In a generic orbit (τ_k, ψ_k), each point can be equally selected as the initial condition. For this analysis, we will select as starting condition (τ_0, ψ_0) the point immediately before the trajectory moves into the left half-plane. Starting from this point and assuming $\psi_0 = M$, with M a positive integer, at the Mth iterate, the trajectory lies on the τ axis ($\psi_M = 0$). The conditions for the existence of an orbit are that at the $2M$th iterate $\tau_{2M} < 0$, and at the $2M+1$th iterate $\tau_{2M+1} \geq 0$. Only if those conditions are satisfied, due to the symmetry of the vector field about the origin, the trajectory from $k = 2M$ to $k = 4M$ will have the same form as the trajectory from $k = 0$ to $k = 2M$, but with opposite directions in the τ and ψ axes. Hence, the point (τ_{4M}, ψ_{4M}) will coincide exactly with the starting point (τ_0, ψ_0), and the trajectory will repeat itself in a closed orbit. These necessary and sufficient conditions are expressed mathematically by the following set of equations:

$$\begin{cases} \tau_0 \geq 0 \\ \tau_1 < 0 \\ \tau_{2M} < 0 \\ \tau_{2M+1} \geq 0 \\ \psi_0 = M. \end{cases} \tag{8}$$

From the considerations above, it follows immediately that M is the radius of the orbit on the ψ axis, and that the orbit itself will have a period equal to $4M$ time steps.

By using the nonlinear map (7), τ_1 can be calculated as

$$\tau_1 = \tau_0 - R\psi_{-D} - \operatorname{sgn}(\tau_{-D}).$$

Under the maximum orbit assumption $2M \geq D$, then $\tau_{-D} \geq 0$ and $\psi_{-D} = M - D$. Hence, $\tau_1 = \tau_0 - R(M-D) - 1$ and the first two equations in (8) can be simplified in

$$0 \leq \tau_0 < R(M-D) + 1.$$

In order to calculate the value of τ_{2M}, the nonlinear map (7) has to be explicitly evaluated. Taking the sum for $k = 0 \ldots i-1$ of both members of the first equation of (7), the value of τ at the ith iterate can be found

$$\tau_i = \tau_0 - R \cdot \sum_{k=0}^{i-1} \psi_{k-D} - \sum_{k=0}^{i-1} \operatorname{sgn}(\tau_{k-D}). \tag{9}$$

By inspection of Fig. 6 the values of ψ and $\operatorname{sgn}(\tau)$ of interest are easily evaluated

$$\psi_{k-D} = \begin{cases} M + k - D, & \text{for } k - D \leq 0 \\ M - k + D, & \text{for } k - D \geq 0 \end{cases}$$
$$\operatorname{sgn}(\tau_{k-D}) = \begin{cases} +1, & \text{for } k - D \leq 0 \\ -1, & \text{for } k - D > 0. \end{cases}$$

By substituting these expression into (9), τ_i can be written explicitly as

$$\tau_i = \begin{cases} \tau_0 - iR\left(M - D + \frac{1-i}{2}\right) - i, & \text{for } i \leq D+1 \\ \tau_0 - R\left[iM - D(D+1) + iD \right. \\ \qquad \left. + \frac{i(1-i)}{2}\right] - 2D - 2 + i, & \text{for } i \geq D+2. \end{cases} \tag{10}$$

In order to compute τ_{2M} and τ_{2M+1}, it is necessary to know which of the two equations in (10) apply. The maximum orbit assumption $(2M \geq D)$ does not help to exclude the first equation. Assuming

$$2M \geq D + 2 \tag{11}$$

the second equation applies always. Note that this is a stronger condition than the maximum orbit assumption. Its validity will be checked *a posteriori*, after the minimum value M for the possible orbits will be derived. Substituting $i = 2M$ and $i = 2M+1$ in the second equation of (10)

$$\tau_{2M} = \tau_0 - R[M(1+2D) - D(D+1)] - 2(D+1-M)$$
$$\tau_{2M+1} = \tau_0 - RD(2M-D) + 2M - 2D - 1.$$

Substituting the above expression for τ_{2M} and τ_{2M+1} into the set of equations (8) and defining the following quantities:

$$\tau_{\min} \triangleq M(2RD - 2) - RD^2 + 2D + 1$$
$$\tau_{\max 1} \triangleq MR + 1 - RD$$
$$\tau_{\max 2} \triangleq M(2RD + R - 2) - RD^2 + 2D + 2 - RD \tag{12}$$

the conditions (8) can be all expressed in the term of τ_0 as

$$\max\{0, \tau_{\min}\} \leq \tau_0 < \min\{\tau_{\max 1}, \tau_{\max 2}\}.$$

This inequality can be interpreted as follows. Given the BBPLL parameters R and D, if there exists at least one positive integer M such that the interval $[\max\{0, \tau_{\min}\}, \min\{\tau_{\max 1}, \tau_{\max 2}\})$ is not empty, then for every value of τ_0 in this interval, an orbit having radius M and visiting the point (τ_0, M) can be built. Thus the necessary

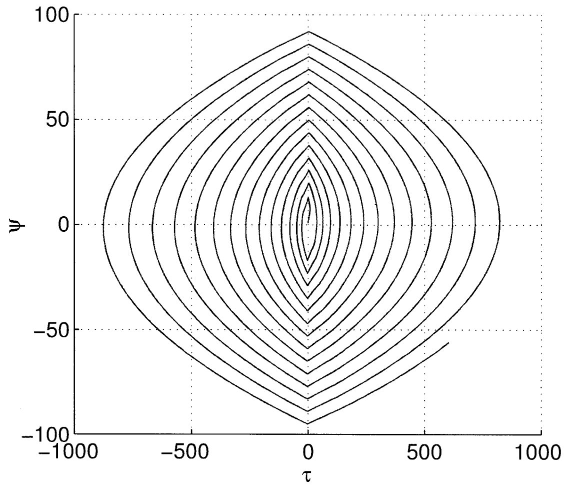

Fig. 7. Diverging trajectory in Region 1, $D = 6, R = 2/(2D-2)$, initial condition $= (0,0)$.

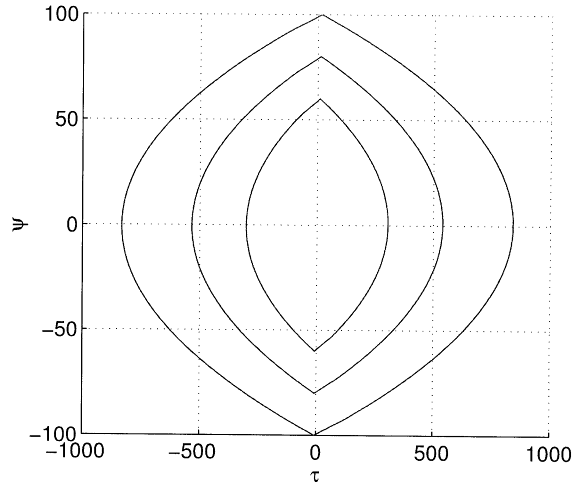

Fig. 8. Three examples of different orbits in Region 2, $D = 6, R = 1/D$.

and sufficient condition for the existence of an orbit for a second-order BBPLL is

$$\exists M \in \mathbb{Z}, M > 0 : \max\{0, \tau_{\min}\} < \min\{\tau_{\max 1}, \tau_{\max 2}\}. \tag{13}$$

From the point of view of the design of a digital BBPLL, it is important to know how the parameters R and D affect the existence of orbits for the system. This can be done by evaluating under which conditions on R and D there exists at least one integer M satisfying (13). The detailed analysis is done in the Appendix. Here only the final results are reported. According to the character of the solutions, three regions can be individuated

$$\begin{aligned}&\text{Region 1}: R \geq \frac{2}{2D-1}\\&\text{Region 2}: \frac{2}{2D+1} \leq R < \frac{2}{2D-1}\\&\text{Region 3}: R < \frac{2}{2D+1}.\end{aligned}$$

In Region 1, there is no M satisfying the condition (13). All trajectories diverge from the origin, meaning that the BBPLL is unstable.

In Region 2, all M greater than a given value $M_{\min}$ satisfies the condition (13). This means that the BBPLL can move in a closed orbit, but the radius M of the orbit is not bounded *a priori*. A BBPLL designed with parameters R and D in this region could lock, but with unknown jitter performances.

In Region 3, the values of M satisfying the condition (13) are all the integer values inside a bounded interval $[M_{\min}, M_{\max}]$, with $M_{\max} < \infty$. That means that the BBPLL locks into a bounded orbit and the worst-case jitter performances can be derived.

Figs. 7 and 8 show some examples of trajectories for BBPLL in Regions 1 and 2. For examples of trajectories in Region 3, see Fig. 5.

For practical applications, only Region 3 is of interest. Indeed, this is the only region which guarantees a finite worst-case value for the radius of the orbits and thus for the jitter of the BBPLL. We will restrict our further analysis on Region 3. The extremes of the interval defining the solutions of (13) are

$$\begin{aligned}M_{\min} &= \frac{D(2+R-RD)}{2+R-2RD}\\M_{\max} &= \frac{(2-RD)(D+1)}{2-R-2RD}.\end{aligned}$$

It can be demonstrated that for all $R \geq 0$ and $D \geq 0$

$$M_{\max} - M_{\min} \geq 1$$

implying that it is always possible to find at least one integer M satisfying the existence condition. In other words, the existence of at least one orbit is guaranteed. It is interesting to note that

$$\begin{aligned}\lim\nolimits_{R\to 0} M_{\min} &= D^+\\\lim\nolimits_{R\to 0} M_{\max} &= (D+1)^+\\\lim\nolimits_{R\to\frac{2}{2D+1}} M_{\min} &< +\infty\\\lim\nolimits_{R\to\frac{2}{2D+1}} M_{\max} &= +\infty\end{aligned}$$

where the superscript + indicates approaching from the right.

The first two equations imply that for R going to zero, the only possible orbits have all the same radius $M = D+1$, leading to a completely predictable jitter performance of the BBPLL. As $M_{\min}$ and $M_{\max}$ are increasing functions of R, it turns out that the minimum value of M for a possible orbit is $D+1$. This guarantees that the assumption (11) holds for each value of D and validates the results of the analytical derivation. The third and fourth equations mean that for R approaching the upper border of Region 3, the range of the possible orbit radius increases indefinitely, leading to more and more uncertainty on the effective jitter performance of the BBPLL.

Fig. 9 illustrates the behavior of $M_{\min}$ and $M_{\max}$ plotted versus R/R_{crit} for $D = 1$ (solid lines) and $D = 4$ (dashed lines), with $R_{\text{crit}} = 2/(2D+1)$. The two vertical lines define the boundaries between the three different regions in the case $D = 1$. From this plot, it can be noted that $D = 1$ and $R/R_{\text{crit}} < 0.4$ guarantees as the only possible solution for the orbit radius the value $M = D+1$. That means that the jitter behavior of the BBPLL can be exactly predicted. For $D = 4$ and $R/R_{\text{crit}} = 0.4$, the only possible solution is $M = D+2$,

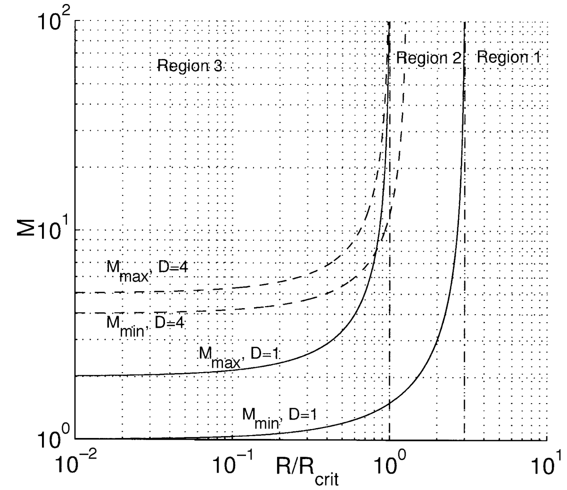

Fig. 9. $M_{\min}$ and $M_{\max}$ plotted versus R/R_{crit} for $D = 1$ (solid lines) and $D = 4$ (dashed). The vertical dash-dot lines show the borders between Regions 1, 2, and 3 for $D = 1$.

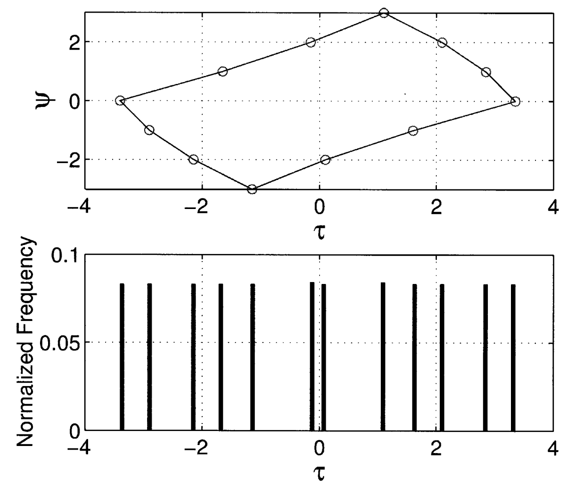

Fig. 10. Orbit with small radius (top) and histogram of τ (bottom).

and the ratio R/R_{crit} must go down to 0.2 in order to have as the only possible solution $M = D + 1$.

C. Evaluations of the Jitter Performances of the BBPLL

Once the possible orbits are known, the jitter performances of the BBPLL can be estimated. It is clear that the jitter will depend on the radius of the orbit that describes the BBPLL behavior in locked condition. The bigger the radius, the bigger will be the jitter. In case that multiple radii are allowed *a priori*, it is not clear which one will be the preferred for the BBPLL orbit. Theoretically, all of them are a possible choice. It could be argued that, from a statistical point of view, the value of M which maximizes the amplitude of the interval specified by (13) has the greatest probability of being selected. However, in designing a system which should guarantee some minimum performances, it is often of more interest to have a reasonable worst-case estimation of the performances themselves. Therefore, a conservative approach will be used here, and the worst-case radius $M = M_{\max}$ will be used to estimate the jitter of the BBPLL. Referring to Fig. 6, the important parameter to calculate is the diameter of the orbit on the τ axis $\tau_{3M} - \tau_M$, which is also equal to the peak-peak jitter τ_{pp}. Due to the symmetry of the orbit about its center $\tau_0 - \tau_M = \tau_{3M} - \tau_{2M}$, hence

$$\tau_{\text{pp}} = \tau_0 + \tau_{2M} - 2\tau_M|_{M=M_{\max}}.$$

Using (10) for calculating τ_M and the already calculated value for τ_{2M}, τ_{pp} as function of R and D can be derived

$$\tau_{\text{pp}} = \frac{(1+D)(2-RD)\left[R^2(3D^2+3D+1)-2R(3D+1)+4\right]}{\left[R(1+2D)-2\right]^2}. \tag{14}$$

Expanding the result in Taylor series around $R = 0$ gives

$$\tau_{\text{pp}} = 2(1+D) + (1+D)R + (1+D)^3R^2 + O(R^3). \tag{15}$$

For $R = 0$, the result of the first-order BBPLL with $x_0 = 0$ is found again. Increasing values of R worsens the peak-peak jitter in a linear way. Therefore, the minimum jitter can be achieved for very small values of R.

In order to evaluate the jitter variance, the probability distribution of τ should be analytically calculated. This task presents some difficulties, and we choose a qualitative alternative, whose validity will be checked *a posteriori* by simulations. Roughly speaking, we can distinguish the possible orbits in two classes: the orbits with a small radius and those with a bigger one. From the considerations above, BBPLLs with small values of D will have orbits of the first class, while BBPLLs with higher D will show orbits of the second class. The orbits with a small radius are made up of few points. Hence, the τ trajectory also consists of few points, and it is reasonable to assume that the distribution of τ among those points will be uniform (see Fig. 10). For this class of orbits

$$\sigma_\tau^2 \approx \frac{\tau_{\text{pp}}^2}{12}. \tag{16}$$

The orbits with a bigger radius are made up of more points. In a first-order approximation, those points can be assumed to be uniformly distributed along the bidimensional trajectory (τ_k, ψ_k). The distribution of the τ values in the interval is equal to the distribution of the projections of those points on the τ axis. From (10), the τ trajectory in one half-plane (for instance, $\tau > 0$) is a quadratic function of the iterate index. In the same half-plane, ψ is a linear function of the iterate index. Thus, neglecting the distortion of the trajectory due to the presence of a delay $D \neq 0$, the trajectory in one half-plane can be seen as a parabolic function $\tau(\psi)$. From the analysis in the previous sections, it is known that the function $\tau(\psi)$ is zero at $\psi = +M$ and $\psi = -M$, and reaches the maximum equal to $\tau_{\text{pp}}/2$ at $\psi = 0$. From these extremes, the function $\tau(\psi)$ can be readily specified

$$\tau(\psi) = \frac{\tau_{\text{pp}}}{2}\left(1 - \frac{\psi^2}{M^2}\right), \quad \psi \in [-M, M]. \tag{17}$$

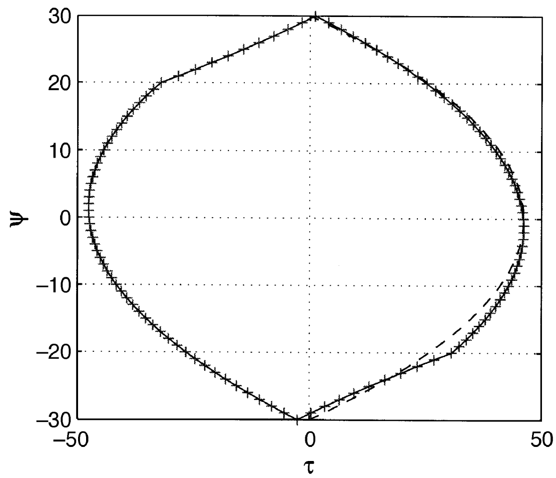

Fig. 11. Orbit with big radius together with the approximation given by (17) shown dashed in the right half-plane $\tau > 0$.

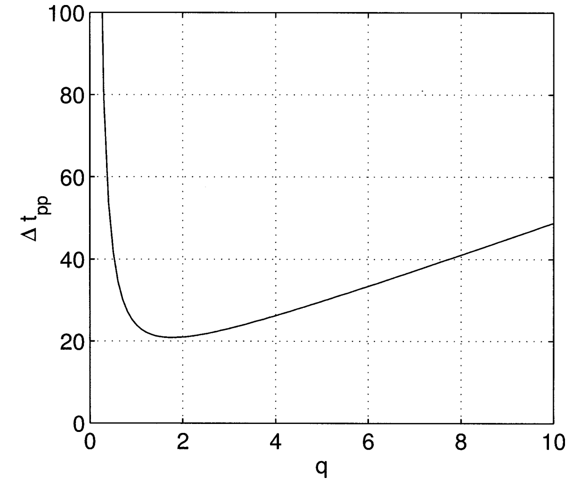

Fig. 12. Peak-peak jitter Δt_{pp} versus $q \triangleq \beta - \alpha(1+2D)/2$ for $\alpha = 1$, $D = 1$, $N = 1$, $K_T = 1$.

Assuming ψ to be a uniformly distributed random variable between $-M$ and M, the jitter variance for this class of orbits can be readily calculated

$$\sigma_\tau^2 = \int_{-M}^{M} \left[\frac{\tau_{\mathrm{pp}}}{2}\left(1 - \frac{\psi^2}{M^2}\right)\right]^2 d\psi = \frac{2}{15}\tau_{\mathrm{pp}}^2. \quad (18)$$

An example of such an orbit is shown in Fig. 11. The dashed line shows the trajectory of τ versus ψ given by (17). The agreement between the actual trajectory and the approximation is good, validating the result (18).

D. Optimization of BBPLL Parameters for Minimum Jitter

In order to evaluate the effects of variations of the BBPLL parameters on the jitter performances, τ has to be denormalized back to Δt. Multiplying both members of (15) by $N\beta K_T$

$$\Delta t_{\mathrm{pp}} = N\beta K_T \left[2(1+D) + (1+D)\left(\frac{\alpha}{\beta}\right) + (1+D)^3\left(\frac{\alpha}{\beta}\right)^2 + O\left(\frac{\alpha^3}{\beta^3}\right)\right].$$

From this expression, jitter minimization can be obtained by minimizing β, D, and the ratio α/β. For fixed β and D, increasing α always worsens the jitter.

A bottom limit to the minimization of β and α is given by the minimum frequency-resolution step of the DCO, which is, in turn, determined by the number of input bits of the DCO itself. However, in most applications, a bottom limit for α comes from different considerations. In general, α, together with the ratio R, determines the transient behavior of the BBPLL in the presence of a phase or frequency step at the input. The bigger the α, the shorter is the locking time of the BBPLL (see [3] for more details). Moreover, clock and data-recovery circuits, which often use a bang–bang architecture [3], must guarantee a given input jitter tolerance. It can be demonstrated that jitter tolerance requires a minimum value of α. Finally, clock-generation circuits for high-speed communications in consumer electronics, where a BBPLL is also a possible solution, often receive as reference a slowly frequency-modulated clock. This technique (called spread-spectrum clock generation) is used to reduce the electromagnetic interferences generated by the circuits clocked by the BBPLL.

For all those considerations, it is of great interest to determine the optimum value of β which minimizes the BBPLL jitter, given fixed α and D values (D is fixed by the implementation). A small value of β (high R) causes the BBPLL to be close to the border between Region 3 and Region 2 (instability border), where the radius of the orbits, and also the jitter, increases dramatically. Bigger values of β (lower R) would stabilize the BBPLL on orbits with small radius. Once β is big enough to allow only one possible orbit, increasing it further will cause the jitter to grow, due to the bigger quantization step in the proportional path of the BBPLL. Thus an optimum value of β exists, which minimizes the jitter.

Define

$$q \triangleq \beta - \frac{\alpha(1+2D)}{2}$$

as the difference between β and the minimum value of β which would lead the BBPLL on the border between Region 3 and Region 2. Using this definition and substituting for β into (14) (after denormalization of τ in Δt), we obtain

$$\Delta t_{\mathrm{pp}} = \frac{NK_T}{4q^2} \cdot \left[(1+D)^4\alpha^3 + 4(1+D)^3\alpha^2 q + 8(1+D)^2\alpha q^2 + 8(1+D)q^3\right].$$

For values of q approaching zero, the peak-peak jitter increases like $1/q^2$, while for $q \to +\infty$, the jitter increases linearly. The plot of the function is shown in Fig. 12 for $\alpha = 1$ and $D = 1$. With standard analytical techniques, the value of q for minimum jitter is found to be at $0.8846461772(1+D)\alpha$. The optimum value of β is then

$$\beta_{\mathrm{opt}} = (1.3846 + 1.8846D)\alpha \quad (19)$$

and, correspondingly, the minimum jitter achievable with a second-order BBPLL is

$$\Delta t_{\mathrm{pp,opt}} = 5.219136251 N\alpha K_T(1+D)^2. \quad (20)$$

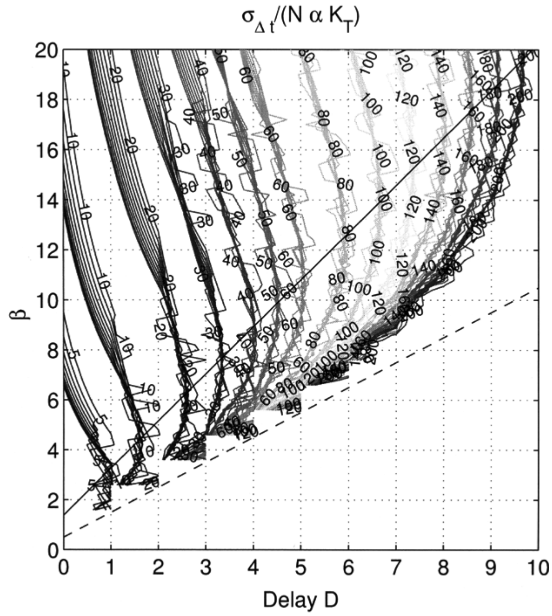

Fig. 13. Contour plot of normalized $\sigma_{\Delta t}$ versus D and β together with β_{opt} (solid) and border between Regions 2 and 3 (dashed).

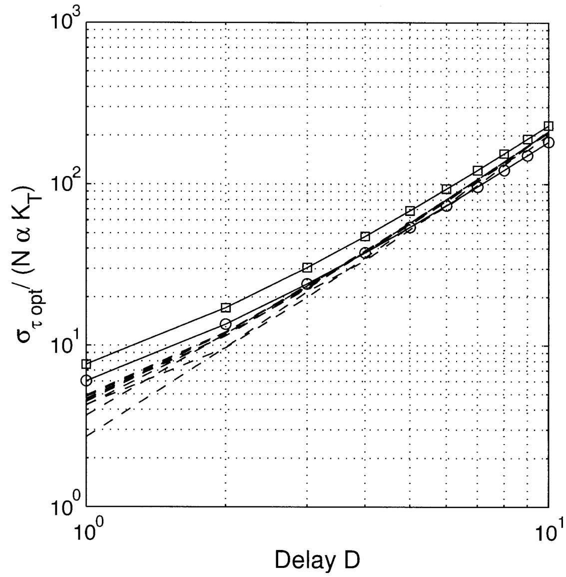

Fig. 14. Minimum normalized $\sigma_{\Delta t}$ values versus D (dashed), together with estimations (16) (circles) and (18) (squares).

VI. Simulation Results

In order to check the validity of (19) and (20), simulations of the nonlinear map (2) have been performed for several values of β and D in Region 3, keeping α constant. The rms jitter $\sigma_{\Delta t}$ has been evaluated, and its value has been normalized to $N\alpha K_T$. The result of the simulations is a surface of normalized rms jitter values versus D and β.

The same simulation has been repeated for different values of input signal $T_r - T_{v0}$ and for several different initial conditions, in order to have the cases of diverging and converging trajectories during the BBPLL locking transient. The result is a set of surfaces of normalized $\sigma_{\Delta t}$ versus D and β. For each of these surfaces, a contour plot has been calculated. Fig. 13 shows the superposition of the contour plots of the set of surfaces, together with the plot of (19) (solid line) and a line representing the border between Region 3 and Region 2 (dashed). It can be seen that (19) matches very well the value of β, leading to the minimum jitter over a wide range of delay D, input signals, and initial conditions.

Fig. 14 shows, for each value of D, the minimum value of $\sigma_{\Delta t}$ on β for all simulations. This plot can be seen as the section of the surfaces of Fig. 13 along the line described by (19). The curves corresponding to (16) (circles) and (18) (squares) are also plotted. The simulations confirm that the estimations for the rms jitter calculated in the previous sections are a reasonable worst case. Further, it can be noted that the estimation done assuming a uniform distribution (16) is not valid any longer for a higher value of D. Instead, for those cases, estimation (18) yields a much better agreement with the simulations results.

VII. Conclusions

In this paper, the nonlinear dynamics of a first- and second-order BBPLL has been analyzed. It has been found that the second-order BBPLL can show three different behaviors: unstable, stable with unbounded orbits, stable with bounded orbits. The effect of the loop delay D on the stability of the BBPLL has been analyzed. It has been proven that in order for the BBPLL to be stable with bounded orbits, the ratio between the integral and proportional constants in the loop filter must be smaller than $2/(2D+1)$. In this case, the optimum value of the proportional constant for minimum jitter has been analytically calculated and the corresponding rms jitter estimated. Simulations were performed, which confirmed the analytical results.

Appendix

In this section, the existence of solutions of (13) as a function of R and D will be evaluated. The three quantities $\tau_{\max 1}$, $\tau_{\max 2}$, and $\tau_{\min}$ are linear functions of M, and can be represented by straight lines in the (M, τ) plane. The notation $\angle\tau_x$ for the angular coefficient of τ_x will be used here.

A. Case $1/D < R < 1/(D-1/2)$, Region 2 Right

In this case, $\angle\tau_{\max 2} > \angle\tau_{\min} > 0$ and $\angle\tau_{\max 1} > \angle\tau_{\min} > 0$. Thus, $\exists M_{\min} < +\infty : \forall M \geq M_{\min}$ condition (13) is satisfied.

B. Case $R > 1/(D-1/2)$, Region 1

In this case, $\angle\tau_{\max 2} > \angle\tau_{\min} > 0$, but $\angle\tau_{\max 1} < \angle\tau_{\min}$. The abscissa of the intersection point between $\tau_{\max 1}$ and $\tau_{\min}$ is

$$M_1 = \frac{RD^2 - RD - 2D}{2RD - R - 2}$$

and the corresponding value on the τ axis is

$$\tau_{\max 1}(M_1) = R\frac{RD^2 - RD - 2D}{2RD - R - 2} + 1 - RD.$$

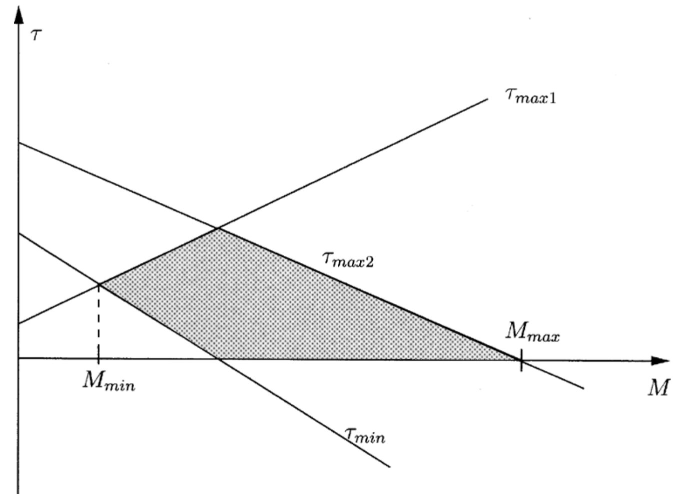

Fig. 15. Illustration of $\tau_{\min}$, $\tau_{\max 1}$, and $\tau_{\max 2}$ for Region 3. The shadowed area are the points satisfying condition (13).

After some algebra

$$\tau_{\max 1}(M_1) = -\frac{(RD-1)^2+1+R}{2RD-R-2}.$$

For $R > 0$, $\tau_{\max 1}(M_1) < 0$, so that the intersection between $\tau_{\max 1}$ and $\tau_{\min}$ occurs on the negative τ half-plane. Thus, in the positive τ half-plane, it is always $\tau_{\max 1} < \tau_{\min}$, therefore, there is no M satisfying (13).

C. Case $1/(D+1/2) < R < 1/D$, Region 2 Left

In this case, $\angle\tau_{\min} < 0$, while $\angle\tau_{\max 1} > 0$ and $\angle\tau_{\max 2} > 0$. Thus, $\exists M_{\min} < +\infty : \forall M \geq M_{\min}$ condition (13) is satisfied.

D. Case $R < 1/(D+1/2)$, Region 3

In this case, $\angle\tau_{\min} < \angle\tau_{\max 2} < 0$, while $\angle\tau_{\max 1} > 0$. Moreover, $0 < \tau_{\min}(0) < \tau_{\max 1}(0) < \tau_{\max 2}(0)$. The situation is illustrated in Fig. 15.

It is clear that condition (13) is satisfied for M in the interval between $M_{\min}$ and $M_{\max}$, where $M_{\min}$ is the intersection between $\tau_{\min}$ and $\tau_{\max 1}$ and $M_{\max}$ is the intersection between the M axis and $\tau_{\max 2}$. Expressions for $M_{\min}$ and $M_{\max}$ are given in the paper body.

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